

Gamma detection for source localisation in targeted alpha therapy

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Abstract

Targeted alpha therapies deliver highly localised dose, but treatment verification depends on knowing whether the radionuclide source remains near the intended site after administration. We simulated an AlphaGlue-like geometry using Geant4, with a thin radioactive glue layer on a water phantom surrounded by six photon-detector panels. Retained, broadly diffused, and locally displaced source states were compared using 50 repetitions of 10,000 primary decays per configuration. Baseline-subtracted gamma-hit profiles distinguished retained, diffused, and localised source distributions. The diffusion fraction was reconstructed from residual profile features with $R^2 = 0.9987$ and a mean absolute error of about one percentage point, while localised-source residuals identified displaced activity with mean position errors of 16–30 mm across the tested localised fractions. These results support decay-chain gamma detection as a candidate observable for post-treatment monitoring and source-retention verification in targeted alpha therapy.

1 Introduction

Targeted alpha therapy is attractive because alpha particles can deliver a large dose over a short range. This locality is also a practical challenge: the delivered dose depends not only on the administered activity, but on whether the radioactive material and its progeny remain near the target, diffuse into adjacent tissue, or are transported through biological processes [2, 3]. The AlphaGlue concept proposes applying alpha-emitting material to a tumour region in a thin glue layer [1]. The activity loaded into the glue is known at production, but confirming the source distribution at treatment time without disturbing delivery remains a non-trivial verification problem.

This work tests whether gamma emissions from the decay chain can provide a theranostic monitoring signal. The study does not model a full clinical detector response; instead, it asks whether idealised detector-entry patterns around a phantom contain enough spatial information to distinguish retained, broadly diffused, and locally displaced source states. The emphasis is on baseline-subtracted profiles, because total gamma-hit yield alone is expected to be a weak localisation observable.

2 Materials and Methods

Photon transport was simulated with the Geant4 toolkit [4]. The model places a 40 mm × 2 mm × 40 mm loaded glue layer on the +y surface of a 20 cm × 20 cm × 150 cm water phantom. Six photon-detector panels surround the phantom at 20 cm from the water surface. Each gamma entry into a detector panel is recorded as one

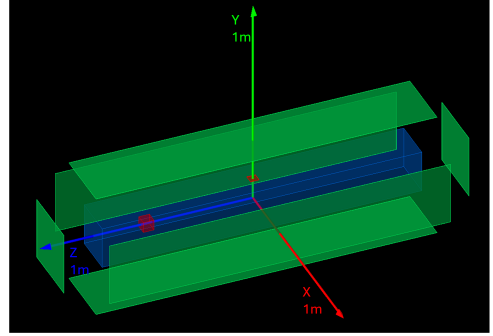


Figure 1: *Simulation geometry with water phantom, surrounding detector panels, and source regions.*

hit and binned by detector panel and longitudinal position.

Seven source states were studied. The baseline case places all activity in the glue layer. Broad-redistribution cases place 10%, 50%, or 90% of the activity diffusely in the water volume, with the remaining activity retained in the glue. Localised-displacement cases place 20%, 40%, or 60% of the activity in a compact displaced water region, again with the remainder retained in the glue. These source states are sensitivity tests for source movement rather than patient-specific biological models.

Each source state was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The retained-source case defines the baseline. Candidate profiles are compared with this baseline using candidate-minus-baseline detector-hit histograms. The run-to-run variation of the baseline repetitions provides the uncertainty scale used for detectability tests.

3 Results

3.1 Baseline-subtracted profiles

The total detector-entry rate changed only weakly between retained and redistributed source states. In the poster analysis, the total detector entry rate changed by only about 3.8% between the retained baseline and the 90% broad-diffusion case. This small global change indicates that total counts alone are insufficient for source localisation.

Spatial residual profiles were much more informative. Broad diffusion widened the residual pattern and changed the balance between the glue-facing and opposite detector panels. Localised displacement produced a shifted excess near the displaced source region. The sign and position of the candidate-minus-baseline residuals therefore encoded whether activity had spread broadly or accumulated in a compact off-applicator region.

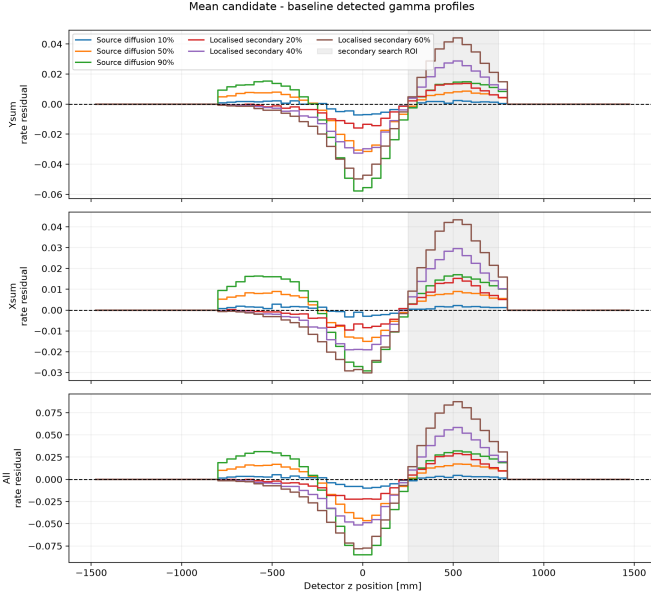


Figure 2: Mean baseline-subtracted detector-hit profiles used as source-state features.

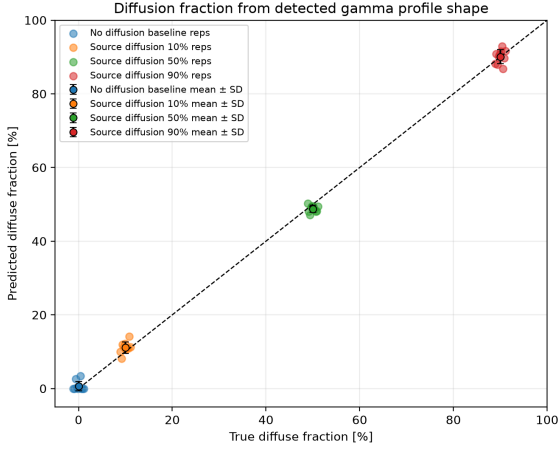


Figure 3: Cross-validated diffusion-fraction prediction from residual profile features.

3.2 Diffusion inference

Residual-shape features were used to estimate the fraction of activity in the broadly diffused component. A regression trained on the retained and broad-diffusion source states reconstructed diffusion fraction with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. The high reconstruction accuracy in this controlled simulation reflects the large change in spatial profile shape compared with the modest change in total detector-entry rate.

3.3 Localised-source detection

A displaced-region region-of-interest metric was used to test for activity outside the expected retained-source region. For localised cases, positive residuals in the secondary-source region provided a source-position estimate. The simulated localised source was centred at $z = 500$ mm. Mean reconstructed positions were approximately $z = 516$ mm, 528 mm, and 530 mm for 20%, 40%, and 60% localised activity, respectively. Across the tested localised fractions, mean position errors were 16–30 mm, with uncertainty decreasing for larger localised

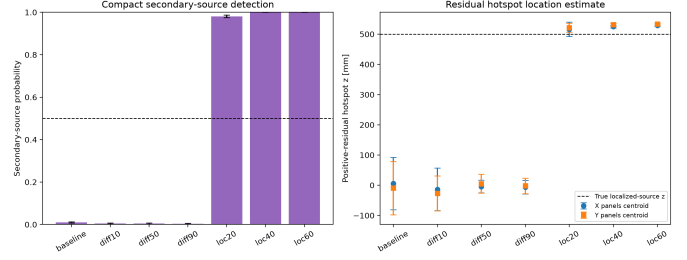


Figure 4: Secondary-source detection and position summary from baseline-subtracted residuals.

fractions.

3.4 Time-window detectability

Detector hits were accumulated in increasing time windows and compared with the retained-source baseline. Detectability was defined using candidate-baseline separation greater than 3σ of the run-to-run baseline variation. In the 50-repetition analysis, 50–90% broad diffusion and 20–60% localised displacement crossed threshold in about 1–2 hours, while the 10% broad-diffusion case required about 6 hours. These results suggest that the residual signal is not only present in integrated profiles, but can emerge on treatment-relevant observation timescales in the simulated geometry.

4 Discussion

The simulation shows that source localisation information is carried primarily by spatial redistribution of decay-chain gamma hits. This is important because the total detector-entry rate changed by only a few percent even for the largest broad-diffusion case. A detector system relying only on total count rate would therefore have limited ability to determine whether activity remained in the applicator region or moved into the phantom. By contrast, baseline-subtracted detector profiles separated retained, diffused, and localised activity states in this idealised model.

The diffusion and localised-source metrics answer complementary questions. The diffusion estimator quantifies broad redistribution using residual shape features calibrated on diffuse source states. The localised-source residual tests for a compact off-applicator excess and estimates its longitudinal position. Reporting both quantities is preferable to interpreting any single profile-width change as diffusion, because a compact displaced source can also broaden or distort the observed profile.

Several limitations remain. The detector panels in this study are idealised hit counters, and the analysis does not yet include detector efficiency, energy response, background, dead time, shielding, acquisition electronics, or patient-specific anatomy. The tested source states are simplified sensitivity cases rather than biologically complete transport models. Future work should connect the activity-normalised residual profiles to measurable count rates in a realistic detector and evaluate robustness under background and clinical geometry assumptions.

5 Conclusion

Baseline-subtracted decay-chain gamma profiles distinguished retained, broadly diffused, and locally displaced AlphaGlue activity in Geant4 simulations. Diffusion fraction was estimated with $R^2 = 0.9987$ and about one percentage-point mean absolute error, while localised-source residuals identified displaced activity with mean position errors of 16–30 mm. These findings support gamma-based post-treatment monitoring as a route to assess source retention and redistribution after targeted alpha therapy, provided future studies confirm the signal under realistic detector-response and patient-scale conditions.

References

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